

DAO Decentralization: Voting-Bloc Entropy, Bribery, and Dark DAOs

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1 Introduction

A Decentralized Autonomous Organization (DAO) is an entity or community that operates based on rules encoded in a smart contract. As the name suggests, a DAO’s governance is decentralized, meaning that it does not rely on a single individual or highly concentrated authority—in contrast to, e.g., a corporation, where a CEO and board of directors make major decisions. Instead, decisions in a DAO are typically made through community votes on proposals. A DAO’s treasury, consisting of crypto assets, also generally resides in its smart contract. The contract enforces adherence to community decisions regarding use of its treasury and also offers operational transparency.

DAOs can serve many goals, including investment (e.g., The DAO [35], Mantle Network [7]), grant distribution (e.g., MolochDAO [8], ResearchDAO [11]), gaming-guild organization (e.g., AvocadoDAO [2], GuildFi [5]) and—as is the case for DAOs with the largest treasuries—ecosystem governance (e.g., Uniswap [13], Lido [6], Arbitrum [1], Optimism Collective [10], MakerDAO [31]).

DAOs of all types are rising rapidly in popularity. At the time of writing (Aug. 2023), the aggregate value across all DAO treasuries exceeds \$20 billion [3], more than double the amount just a year ago.

DAOs today vary considerably in their *true* degree of decentralization. Most have their own associated crypto assets (or “tokens”) and weigh voting power by token holdings. It is common for vote outcomes to be determined by a small set of “whales”—a colloquial term used to denote the largest token holders. Such centralization, as well as low voting participation, are a pervasive source of concern in DAO communities. Vulnerability to centralization has even led to plundering of DAO treasuries [32].

A number of works have sought to recommend ways to improve DAO decentralization. But first it’s necessary to be able to *measure* it in a way that is reflective of a broad set of real-world concerns. That requirement is the starting point for our work in this paper.

1.1 Measuring DAO decentralization

A common basis for evaluating decentralization in DAOs and other blockchain settings is *token ownership*, specifically the distribution of assets and consequently voting rights among

participants [38]. Informally, concentration of a large fraction of tokens in a small number of hands—and thus the ability of a small group to determine voting outcomes—is indicative of strong centralization. More widespread distribution, conversely, suggests decentralization.

Entropy is one popular metric for measuring decentralization in the distribution of token ownership in a DAO.¹ For a set of addresses $A = \{a_1, \dots, a_n\}$, where address a_i holds t_i tokens and $T = \sum_{i=1}^n t_i$:

$$\text{entropy}(A) \triangleq - \sum_{i=1}^n \frac{t_i}{T} \log \left(\frac{t_i}{T} \right).$$

Low entropy corresponds to a high degree of asset concentration and thus strong centralization. High entropy implies the opposite. Other popular decentralization metrics, e.g., the Gini coefficient [25] and the Nakamoto coefficient [9], are related to various notions of entropy.

Token ownership distribution alone, however, has serious shortcomings as a decentralization metric. To begin with, it is visible on chain only in terms of per-address holdings, not control by real-world individuals. Thus, for instance, an individual who holds 51% of tokens in a DAO, but spreads them among a large number of addresses could create an appearance of decentralization while having majority control.

Even if tokens are held by distinct entities, a notion put forward in, e.g., [28], those entities may have aligned interests and act in concert—a form of centralization. The following examples illustrate cases in which a DAO may be strongly centralized, *even if token ownership appears to imply strong decentralization*.

Example 1 (Low participation / apathy). Lack of participation in DAO governance votes is widespread in practice [20] and induces a form of centralization. Consider, for example, a DAO that requires a quorum of 50% participation for a vote to be ratified. Suppose 50% of voters do not cast votes and voters other than whales vote 2:1 in favor of the proposal. Whales with just 12.6% of all tokens can swing the vote and cause the proposal to be rejected.

Example 2 (Social pressure). Interviews with DAO participants have revealed a tendency to vote in alignment with influential community members for the sake of preserving reputation [38], as individual votes are today usually publicly observable. This phenomenon exerts a centralizing force, as it aligns votes around the choices of a small set of participants. (This problem is similar to the notion of “herding” in classical voting theory [26, 15].)

Example 3 (Bribery / vote-buying). Bribery—specifically, *vote-buying*—has been a long-standing concern of DAO organizers [18]. It has a centralizing effect, as it aligns voters around a choice dictated by the briber.

Recognizing that token-ownership alone doesn’t give a full picture of decentralization, researchers have explored broader notions. Most notably, Sharma et al. [38] have considered entropy measures limited to those voters who participate in votes and have also explored graph-based representations of voting patterns (degree centralization, degree assortativity,

¹Entropy is typically defined over a random variable. A token ownership distribution may be viewed as a random variable for an experiment where a token is selected uniformly at random and its owner is output.

etc.). Token-ownership distribution among voting participants fails to capture important issues, such as those in Examples 2 and 3, however, and it's unclear how to interpret graph-based metrics.

With no consensus in the community about how to measure DAO decentralization today, there is a lack of principled guidance on ways to improve DAO decentralization and to combat threats to decentralization, such as vote-buying.

1.2 Voting-Bloc Entropy (VBE)

We introduce a decentralization metric tailored to DAO governance called *Voting-Bloc Entropy* (VBE). VBE is based on a foundational principle, that voters with closely aligned interests across elections are a centralizing force. Expressed differently, the key idea in VBE is to *define centralization as the existence of large voting blocs*.

Formally, we express this principle in terms of the *utility functions* of DAO participants, i.e., quantification of the gain or loss associated with voting outcomes. For a given set of elections, a voting bloc is a cluster of voters whose utility functions are similar over outcomes.

VBE then, measures entropy over voting blocs based on utility functions—rather than over individual token holdings. The result is a broad concept that captures the centralization embodied in all of our examples above. VBE is in fact a framework: It allows different notions of clustering and entropy to be plugged in.

We stress that VBE is a theoretical metric: It cannot be measured *directly*, since users do not typically express (or often even know) their utility functions explicitly. But VBE provides an important basis for reasoning about decentralization, and does lend itself to *indirect* measurement.

VBE Implications: We use VBE to prove a number of theoretical results showing how various practices tend to increase or decrease DAO decentralization. (We prove these results relative to particular notions of clustering and entropy.)

Our main results are as follows:

- **Apathy / inactivity whale:** A large population of apathetic, i.e., non-voting DAO participants, is a centralizing force (Theorem 3.2).
- **Delegation:** Given an inactivity whale of large size relative to delegates, delegation tends (perhaps counterintuitively) to increase decentralization (Theorem 3.3).
- **Bribery:** Bribery and decentralization are closely related in the context of DAO governance. The act of bribery decreases decentralization (Theorem 3.6). Additionally, as the decentralization of a DAO rises, so does the the cost of bribery—and vice versa (Theorem 3.7).

We additionally prove results relating to social pressure and privacy (Theorem 3.4), and owning multiple accounts (Theorem 3.1).

1.3 Dark DAOs

Our results on bribery and decentralization (Theorems 3.6 and 3.7) show that as decentralization increases, bribery can only be successful if it operates on a large scale. This observation raises a critical followup question: Is large-scale bribery a realistic future threat to DAOs?

One mechanism postulated for systemic DAO-voting bribery is a *Dark DAO*—a DAO designed to corrupt votes in other DAOs. A Dark DAO was originally defined as “a decentralized cartel that buys on-chain votes opaquely.” [21]. Here, “opaquely” means that participation in the Dark DAO is confidential. A Dark DAO must also ensure correct execution of a bribery scheme, i.e., bribes are paid if and only if they vote as directed.

To date, the feasibility of a fully functional Dark DAO has yet to be demonstrated. In this work, we present a Dark DAO prototype that leverages the confidentiality enforced by trusted hardware (specifically Intel SGX) in the Oasis Sapphire blockchain² to facilitate vote-buying in DAOs on Ethereum—the most popular blockchain for DAOs today. Our work demonstrates that Dark DAOs are indeed a realistic future threat. We also briefly discuss possible mitigations.

1.4 Contributions

In summary, our contributions in this work are:

- **Voting-Bloc Entropy (VBE):** We introduce VBE, a new metric for DAO decentralization that generalizes prior metrics and addresses a number of their shortcomings (Section 2).
- **Theoretical results:** Using VBE, we prove a range of results about how various DAO practices and design choices impact decentralization (Section 3).
- **Dark DAOs:** Our theoretical results highlight risks of systemic bribery—through Dark DAOs—despite high decentralization in the target DAOs. We implement a complete Dark DAO to show that these risks are a realistic long-term concern (Sections 4 and 5).
- **Practical guidance:** Based on our theoretical and experimental results, we present concrete points of practical guidance for DAO design and deployment around issues including delegation, voting privacy, voting-slate composition, decentralized identity, and more (Section 6). We summarize this guidance in Table 1.

We conclude with some open research questions in Section 7.

2 Voting-Bloc Entropy (VBE)

In this section, we define *Voting-Bloc Entropy* (VBE), which sidesteps the aforementioned limitations of prior metrics. It does so by normalizing token holdings based on voters’ utility functions.

²<https://github.com/oasisprotocol/sapphire-paratime>

Intuition. The key idea behind our definition is to reason about centralization with respect to the tokens held by *groups of DAO members with aligned interests*, instead of with respect to individual members. That is, instead of measuring the distribution of tokens across individual addresses, we focus instead on how tokens are distributed across blocs of voters with the same incentives, which are functionally acting as a single entity. Looking ahead, we formalize the notion of “aligned interests” by considering the DAO members’ *utility functions* across elections.

Aggregating voters based on utility functions allows us to capture the rich interactions and relationships between players in the system, all of which play a role in understanding the true degree of decentralization of a DAO, as discussed in Section 1.1. Indeed, the limitations highlighted there are captured by considering voters with the same utility functions as a single entity; we discuss this extensively in Section 3.

DAO abstraction. We now introduce the notation and formalism that our definition and theorems rely on.

Let $\mathcal{P} = \{P_1, \dots, P_n\}$ be the set of token holders in a system, for which we use **tokens**: $\mathcal{P} \rightarrow \mathbb{R}^+$ to define the number of tokens held by each $P \in \mathcal{P}$. (We will often overload this notation, and input a *set* of accounts to **tokens** instead, by which we mean the total tokens held across all accounts in the set). These token holders participate in a set of (binary) elections $E = \{e_1, e_2, \dots, e_m\}$, where we denote by $\text{vote}_P: E \rightarrow \{\text{true}, \text{false}, \perp\}$ player P ’s vote in election e ; $\perp$ indicates that P abstained from voting in e . We define $\text{util}_P: E \times \{\text{true}, \text{false}\} \rightarrow \mathbb{R}$ to be the monetary utility of outcome **true** and **false** in e to player P , where we make the simplifying assumption that $\text{util}_P(e, \text{true}) = -\text{util}_P(e, \text{false})$. Player P ’s total utility across all elections E is represented by a vector $U_{E,P} := (\text{util}_P(e_i, \text{true}))_{i \in [m]} \in \mathbb{R}^m$; we denote by $U_{E,\mathcal{P}}$ all players’ utilities, i.e., $U_{E,\mathcal{P}} := (U_{E,P})_{P \in \mathcal{P}}$.

Token holders often have low stakes in the elections, resulting in lack of interest or abstaining from voting altogether. More formally, we say that player P is ϵ -*apathetic* in election e if and only if $|\text{util}_P(e, \text{true})| \leq \epsilon$. If the system supports vote delegation, especially as a means to combat apathetic voters, players may delegate their tokens to others, who cast a single vote on behalf of all the tokens they now hold.

Lastly, we define $\text{bribe}_P: E \times \{\text{true}, \text{false}\} \rightarrow \mathbb{R}$ to be such that it is possible for player P to achieve an outcome of **true** (resp., **false**) in a particular election e via bribery for any expenditure greater than $\text{bribe}_P(e, \text{true})$ (resp., $\text{bribe}_P(e, \text{false})$). Note that we make the simplifying assumption that bribery costs are independent across elections. We assume that bribing P to flip its vote from **true** to **false** (respectively, **false** to **true**) costs $\max(\text{util}_P(e, \text{true}) + \epsilon, 0)$ (respectively, $\max(\text{util}_P(e, \text{true}) + \epsilon, 0)$), for some constant ϵ . Successful bribery to achieve an outcome of **true** in e means flipping enough votes to ensure a majority vote by token weight for **true** in e (and vice versa for **false**), i.e., ensuring

$$\sum_{P \in \mathcal{P}} \left(\text{tokens}(P) \mid \text{vote}_P(e) = \text{true} \right) > \frac{\sum_{P \in \mathcal{P}} \text{tokens}(P)}{2}.$$

We stress that the equation above represents the threshold to *ensure* the desired outcome in an election, and not just to win it. Albeit related, these notions are not equivalent; the

former implies the latter, but not vice-versa. In particular, winning an election is a function of the number of votes cast for the undesired option, whereas ensuring an outcome is agnostic to this.

2.1 Framework for VBE

We define VBE in this section. To do so, we introduce an abstract framework that is parameterized by: (1) a clustering metric, and (2) an entropy notion, which are the two key ingredients that underpin our definition.

Clustering. We let $C: U_{E,\mathcal{P}} \times U_{E,\mathcal{P}} \rightarrow \{0,1\}$ be a clustering function that outputs 1 if the utilities of two players are “aligned” across all elections E , and 0 otherwise. Our definition of VBE is agnostic to a specific clustering algorithm, and instead only assumes that C specifies an equivalence relation $\sim_C$ on the set $\mathcal{P}$, such that $P_i \sim_C P_j$ if and only if $C(U_{E,P_i}, U_{E,P_j}) = 1$. That is, C partitions $\mathcal{P}$ into classes of players with aligned utility functions across elections. Following standard notation, we denote the set of all classes by $\mathcal{P}/\sim_C$, and the class P belongs to by $[P]$.

Entropy. We denote by F some entropy notion over the distribution of tokens across *sets* of accounts.³ If the sets are comprised of single entities, this is equivalent to entropy over individual accounts, as defined in prior work [38].

We are now ready to define VBE. Intuitively, our definition says that a DAO is more decentralized if the distribution of tokens across the abstract voting entities specified by $\sim_C$ has high entropy according to F . More concretely,

Definition 1 (Voting-Bloc Entropy). For a set of elections E , a set of players $\mathcal{P}$ with corresponding utilities $U_{E,\mathcal{P}}$, a clustering metric C , and an entropy function F , we define *Voting-Bloc Entropy* (VBE) to be:

$$\text{VBE}_{C,F}(E, \mathcal{P}, U_{E,\mathcal{P}}) := F(\mathcal{P}/\sim_C). \quad (1)$$

Note that computing an analogous metric over account balances alone gives an upper bound on VBE. That is, let C_{solo} be the clustering metric that partitions $\mathcal{P}$ into singleton sets, i.e., $C_{\text{solo}}(U_{E,P_i}, U_{E,P_j}) := 1 \iff i = j$. Then, for a fixed E , $\mathcal{P}$, and F , it follows that, for any clustering metric C ,

$$\text{VBE}_{C_{\text{solo}},F}(E, \mathcal{P}, U_{E,\mathcal{P}}) \geq \text{VBE}_{C,F}(E, \mathcal{P}, U_{E,\mathcal{P}}).$$

2.2 Instantiation of VBE

There are various concrete algorithms with which one can instantiate our VBE framework. We propose one such example in this section, which we use throughout the rest of the paper. The advantages and disadvantages of this variant, and VBE in general, are discussed in Section 2.3.

³Entropy is formally defined over a random variable, but we are overloading notation to think of the mapping between sets of accounts and their respective cumulative token balances as the probability mass function of a random variable.

Clustering. We define ϵ -threshold ordinal clustering (ϵ -TOC) as follows:

$$C_\epsilon(U_{E,P_i}, U_{E,P_j}) := \begin{cases} 1 & \text{if } \forall k \in [m], \left(\text{sgn}(U_{E,P_i}[k]) = \text{sgn}(U_{E,P_j}[k]) \right) \vee \left(|U_{E,P_i}[k]|, |U_{E,P_j}[k]| \leq \epsilon \right) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

More simply, ϵ -TOC clusters together token holders who have the same preferred outcome across all elections, regardless of how strong this preference is. That is, clusters correspond to token holders whose utility functions are ordinally equivalent. Even though a more granular metric could create clusters based on cardinal utility, we deem ordinal equivalence to be indicative of aligned preferences.

ϵ -TOC creates an additional cluster corresponding to all apathetic voters, i.e., those whose utilities are close to 0. These voters have aligned preferences, namely, little to no interest in election outcomes.

Entropy. In this work, we use *min-entropy* as our entropy notion. That is, for a set of sets of addresses A with a total of T tokens held across all individual accounts,

$$F_{\min}(A) := \log_2 \left(\frac{\max_{A' \in A} \text{tokens}(A')}{T} \right).$$

Our entropy notion thus measures the amount of “information” in the largest voting bloc by token holdings. As we discuss later on, more granular entropy notions result in more detailed analysis (at the cost of being more difficult to measure in practice), as these may capture the information in other voting blocs beyond the largest.

Putting the two together, we thus get a concrete instantiation of the framework:

$$\text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E,\mathcal{P}}) := F_{\min}(\mathcal{P} / \sim_{C_\epsilon}) = -\log_2 \left(\frac{\max_{\mathcal{P}' \in \mathcal{P} / \sim_{C_\epsilon}} \text{tokens}(\mathcal{P}')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right). \quad (3)$$

2.3 Limitations of VBE

Formal model. Our DAO abstraction from Section 2, which underpins VBE, makes several assumptions, which need not be true in practice. For example, we only consider binary elections, whereas some DAO proposals often involve multiple options, e.g., the Optimism election to fund public goods, where votes explicitly expressed the allocation of funds to different organizations. We note, however, that our model can be used to analyze arbitrary elections, as this would simply involve an appropriate clustering metric where there are multiple options that “aligned” voters can choose from.

A more important limitation is the fact that we assume token holdings remain constant across elections. For simplicity and clarity of presentation, we deem this to be sufficient due to the conceptual nature of our theoretical results. Further, our model can easily be reframed to assume variable token holdings. For example, $\text{tokens}(P)$ can be defined to be the maximum number of tokens held by P at any point across all elections E .

Measuring VBE. VBE is a theoretical notion and cannot be directly measured in practice. Without considering a concrete instantiation of VBE, it serves mostly as a conceptual tool to reason about how certain decisions or events may impact the decentralization of a DAO. To derive concrete metrics and analyze the real-world behavior of a system, one must instead initialize our framework with an appropriate clustering metric and entropy function. This has two main limitations: (1) picking a clustering and entropy function may not be trivial; and (2) we cannot directly compare the metrics from different instantiations of VBE.

Deciding on clustering and entropy functions is primarily a function of the available data. In general, more granular notions are more informative. For example, Shannon entropy yields more nuanced results than min-entropy, and a clustering metric based on cardinal utilities will result in blocs of players that are more closely aligned. However, such functions may require data that is not easy to gather, or may not even be computable at all. For instance, a clustering metric derived from the ordinal value of the players’ utility requires knowing (or estimating) the specific monetary gain that each player derives from the outcome of an election, which need not even be exclusively a function of on-chain data. As such, there is a trade-off between how informative VBE is, and how easy it is to compute.

Two instances of VBE are not directly comparable. That is, in order to reason about the relative level of decentralization between two DAOs, or how decentralization has fluctuated over time for a single DAO, the same variant of VBE must be used. Even though this is an important limitation, in practice we expect there to be a handful of industry-standard parameters agreed upon by the DAO community.

Limitations of $\text{VBE}_{C_e, \min}$. In addition to the general limitations described above, each variant of VBE may have additional shortcomings. In the case of $\text{VBE}_{C_e, \min}$, we lose most of the information provided by all voting blocs except the largest one, since min-entropy is only a function of the latter. This does not imply that the analysis is not accurate, as all subsequent blocs are strictly smaller than the one our definition focuses on, but rather that other entropy notions may be more insightful; indeed, min-entropy is always less than or equal to, e.g., Shannon entropy or max-entropy [41].

Our clustering metric, ϵ -TOC, is also quite strict, as it does not reason about voters who are aligned in most (but not all) elections. One could instead consider, for example, a generalization ϵ -threshold ordinal clustering that is parametrized by the fraction of elections two players must agree on to be considered part of the same cluster. We opted for the more simple variant in this work, as it serves as a proof-of-concept for our theoretical results; more general clustering metrics would yield the same conceptual conclusions, while making the theorem statements and proofs more opaque with orthogonal mathematical details.

3 Implications of VBE: Theoretical Results

We now present a variety of theoretical results implied by VBE. These results show how, unlike prior work, VBE reflects many of the subtle issues that impact decentralization in a DAO—such as those described in Section 1.1—and thus more accurately supports the goals of a DAO.

3.1 Owning multiple accounts

The most straightforward example of the opacity of token ownership across individual accounts is the fact that a single entity may own multiple addresses. Even though it may seem like these tokens are diversified across the accounts, in practice, they are all under the control of one entity. Unlike prior work, our definition is able to capture this nuance, since these accounts would indeed be considered a single voting bloc (we make the simplifying assumption that an individual’s utility function is the same across all her accounts; this need not always be true). We formalize this below.

Theorem 3.1. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players’s utilities across the elections. Assume that a single entity instead controls the accounts associated with players $\hat{\mathcal{P}} \subseteq \mathcal{P}$. We denote the new set of utilities by $\hat{U}_{E,\mathcal{P}}$. Then, it follows that

$$\text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E,\mathcal{P}}) \geq \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, \hat{U}_{E,\mathcal{P}}). \quad (4)$$

Proof. We first note that all voters in $\hat{\mathcal{P}}$ clearly belong to the same voting bloc B' , given that they are owned by a single entity. (Recall that we are assuming that this entity has the same utility function across all her accounts.)

Let B'' be the largest voting bloc by token holdings according to $U_{E,\mathcal{P}}$. Note that $\text{tokens}(B'') \leq \text{tokens}(B')$: either the “new” voting bloc that results from aggregating all of $\hat{\mathcal{P}}$ is now the largest bloc, or the same bloc is the largest in both instances. It follows then that

$$\begin{aligned} & \text{tokens}(B'') \leq \text{tokens}(B') \\ \implies & \frac{\text{tokens}(B'')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \leq \frac{\text{tokens}(B')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \\ \implies & -\log_2 \left(\frac{\text{tokens}(B'')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) \geq -\log_2 \left(\frac{\text{tokens}(B')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) \\ \implies & \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E,\mathcal{P}}) \geq \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, \hat{U}_{E,\mathcal{P}}) \end{aligned}$$

as desired. $\square$

If $\hat{\mathcal{P}}$ represents a majority by token holdings, then the new VBE is strictly larger; equality follows otherwise. This same logic will be true in the subsequent theorems, where VBE increases if the “new” voting bloc holds a majority of tokens.

3.2 Apathy

A DAO whose voters are apathetic, i.e., not interested in the direction of the community, is clearly not aligned with the goals of a DAO. Distribution of tokens is irrelevant if individuals abstain to vote or delegate their votes, as elections are narrowed squarely to the set of more invested stakeholders. Our definition captures this fact. Intuitively, apathetic voters all have

similar utility functions, which reflect their lack of stake in the elections. VBE groups all of these players within the same voting clique, due to their aligned utilities, thus concluding that the system is more centralized. We formalize this below.

Theorem 3.2. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players's utilities across the elections. Assume that players $\hat{\mathcal{P}} \subseteq \mathcal{P}$ by token holdings are instead ϵ -apathetic, i.e.,

$$\forall P \in \hat{\mathcal{P}} \forall e \in E, |\text{util}_P(e, \text{true})| \leq \epsilon.$$

The utilities of all players in $\mathcal{P} - \hat{\mathcal{P}}$ are unchanged. We denote the new set of utilities by $\hat{U}_{E,\mathcal{P}}$. Then, it follows that

$$\text{VBE}_{C_{2\epsilon}, \min}(E, \mathcal{P}, U_{E,\mathcal{P}}) \geq \text{VBE}_{C_{2\epsilon}, \min}(E, \mathcal{P}, \hat{U}_{E,\mathcal{P}}). \quad (5)$$

Proof. We first note that all apathetic voters belong to the same voting bloc, according to ϵ -TOC: for all $P_i, P_j \in \hat{\mathcal{P}}$ and $e \in E$, it follows that

$$\begin{aligned} & |\text{util}_{P_i}(e, \text{true}) - \text{util}_{P_j}(e, \text{true})| \leq |\text{util}_{P_i}(e, \text{true})| + |\text{util}_{P_j}(e, \text{true})| \\ \implies & |\text{util}_{P_i}(e, \text{true}) - \text{util}_{P_j}(e, \text{true})| \leq 2\epsilon. \end{aligned}$$

Where the implication follows from the definition of ϵ -apathetic. According to $C_{2\epsilon}$, then, all $P \in \hat{\mathcal{P}}$ belong to the same voting bloc B' for all elections in E , since their utilities are always at most 2ϵ apart from each other. The rest of the proof then proceeds as in Theorem 3.1. $\square$

We refer to the bloc of apathetic in a DAO, i.e., non-voting token holders, as the *inactivity whale*. This term reflects the collective and potentially systemically important inactive behavior of this group.

3.3 Delegation

Common sense would suggest that delegation leads to a more centralized system: tokens that were originally held by a large set of players, are instead under the control of the (smaller) set of delegates. As we prove formally below, however, this intuition is not accurate: delegation actually tends to make a DAO more decentralized. This insight is clear once we reason about voting blocs instead of individual addresses: before delegation, the tokens are all held by a *single* voting bloc, namely, the inactivity whale. Delegation then diversifies the tokens held by this “whale”, and distributes them amongst a set of voting blocs (the delegates). Assuming that size of the inactivity whale is larger than every delegate's total tokens—which tends to be true in practice[20]—the system is now more decentralized.

Theorem 3.3. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players's utilities across the elections. Assume that a majority of players $\hat{\mathcal{P}} \subseteq \mathcal{P}$ by token holdings, who are ϵ -apathetic, delegate their votes to a smaller set of delegates $D \subset \mathcal{P}$. We denote the new set of utilities by $\hat{U}_{E,\mathcal{P}}$ and, for all $d \in D$, the

total tokens now controlled by $[d]$ by $n_{[d]}$ (i.e., $\text{tokens}([d])$ plus all tokens delegated to $[d]$). Then, it follows that,

$$\forall d \in D, \text{tokens}(\hat{P}) > n_{[d]} \iff \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, \hat{U}_{E, \mathcal{P}}) > \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E, \mathcal{P}}). \quad (6)$$

Proof. As discussed in the proof of Theorem 3.2, all players in $\hat{P}$ belong to the same voting bloc B' for all elections in E , i.e., the inactivity whale. Let B'' be the largest voting bloc by token holdings after delegation; note that B'' may potentially be one of the delegates themselves. We now discuss each direction separately.

($\implies$) First note that, by assumption, $\text{tokens}(B') > n_{[d]}$. Then, since B' represents a majority of token holdings a priori, the blocs formed by non-delegates are also smaller than B' . Thus, irrespective of whether delegates are part of B'' or not, $\text{tokens}(B'') > \text{tokens}(B')$. The rest of the proof then proceeds as in Theorem 3.1.

($\impliedby$) Since VBE increased, it must be the case that B'' is smaller than B' . More concretely, by the definition of VBE we get that

$$\begin{aligned} & \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, \hat{U}_{E, \mathcal{P}}) > \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E, \mathcal{P}}) \\ \implies & -\log_2 \left(\frac{\text{tokens}(B'')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) > -\log_2 \left(\frac{\text{tokens}(B')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) \\ \implies & \text{tokens}(B'') < \text{tokens}(B') \end{aligned}$$

Since, by definition, $n_{[d]} \leq \text{tokens}(B'')$, it follows then that $n_{[d]} < \text{tokens}(B')$, as desired. $\square$

The intuition behind this result is that, as long as the delegates are not “too big”, delegation actually aids in decentralization. Conversely, if some delegate is a whale, or gets delegated an overwhelming majority of tokens, then the system becomes more centralized. This concept is represented by the following corollary, which is simply the contrapositive of Theorem 3.3

Corollary 1. For some $d \in D$, $n_{[d]} > \text{tokens}(\hat{P})$ if and only if $\text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, \hat{U}_{E, \mathcal{P}}) > \text{VBE}_{C_\epsilon, \min}(E, \mathcal{P}, U_{E, \mathcal{P}})$.

In practice, it is much more common for delegates to be small relative to the inactivity whale. That is, delegation is most useful in cases where apathy is high. However, there are certain scenarios where this need not be true. For example,

3.4 Social Pressure

A core goal of DAOs—and any democratic system for that matter—is for token holders to vote according to their true preferences. In practice, however, many DAOs fail to meet this goal: when the votes are publicly observable, token holders have reported feeling influenced to vote a certain way, often in alignment with influential community members, in order to thwart the reputational risks associated with opposing popular viewpoints [38]. In such

a case, the negative monetary utility of harming their reputation outweighs the positive monetary utility of their desired outcome in a vacuum.

Social pressure leads to more centralization, as votes are converging on one outcome. Token distribution alone, however, does not show this. Indeed, a system where tokens are distributed evenly, but all players vote for the same outcome due to social pressure, would be deemed optimally decentralized according to such metrics. VBE, on the other hand, does conclude that reputational risks lead to more centralization, as it aligns the utilities of the players towards the socially preferred outcome.

Theorem 3.4. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players’s utilities across the elections. Assume that a subset of players $\hat{\mathcal{P}} \subseteq \mathcal{P}$ feel social pressure towards voting for, without loss of generality, **true**. That is, for all $P \in \hat{\mathcal{P}}$ and $e \in E$, the monetary reputational cost is greater than or equal to $\max(\text{util}_P(e, \text{false}) + \epsilon, 0)$ for some small constant ϵ .

The utilities of all players in $\mathcal{P} - \hat{\mathcal{P}}$ are unchanged. We denote the new set of utilities by $\hat{U}_{E,\mathcal{P}}$. Then, it follows that

$$\text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U_{E,\mathcal{P}}) \geq \text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, \hat{U}_{E,\mathcal{P}}) \quad (7)$$

Proof. Note that all voters in $\hat{\mathcal{P}}$ belong to the same voting bloc. For every $P \in \hat{\mathcal{P}}$, $U_{E,P}$ will consist of only positive values: either P preferred an outcome of **true** to begin with, or their monetary utility of **true** is now ϵ , due to social pressure. Thus, since $\text{sgn}(U_{E,P}[k]) = 1$ for all $k \in [m]$, all of $\hat{\mathcal{P}}$ consists of a single voting bloc according to ϵ -TOC. The rest of the proof then proceeds as in Theorem 3.1. $\square$

An interesting corollary of this theorem is that privacy increases the decentralization of a system, as it serves as a “mitigation” to social pressure. (Informally, we can show that privacy increases $\text{VBE}_{C_{\epsilon},\min}$ via a proof by contradiction that just directly invokes Theorem 3.4.)

3.5 Voting slates

Grouping together various elections into a lesser number of (more general) elections—so-called “voting slates”—is in opposition with decentralization: decision-making is forced to be more diluted, thus decreasing the relative impact of each voter into the underlying proposals. Thus, voting slates “factor out” differences in the viewpoints of individuals, yielding more homogenous utilities. For example, two players may disagree in many of the individual proposals of the slates but agree on a few of the more important ones, resulting in them casting the same vote.

Theorem 3.5. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players’s utilities across the elections. Assume that all elections E are grouped together, in some way, to form a smaller set of elections $\hat{E}$. We denote the new set of utilities by $U_{\hat{E},\mathcal{P}}$. Then, it follows that

$$\text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U_{E,\mathcal{P}}) > \text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U_{\hat{E},\mathcal{P}}) \quad (8)$$

Proof. We argue this informally, as the proof is very similar to that of the prior theorems. Let B' and B'' be the largest voting blocs by token holdings according to $U_{E,\mathcal{P}}$ and $U_{\hat{E},\mathcal{P}}$, respectively. Then, note that $|B''| \geq |B'|$. First, every player in B' is also in B'' : if two players agree on all individual elections in E , then they will still agree in the case where elections are simply grouped together. Conversely, assume that two players from B'' both desire an outcome of, without loss of generality, **true** in some election in $\hat{E}$. This need not imply, however, that they have aligned utilities in all the elections from E comprised therein, but rather that their total added utilities are both positive.

Since the grouping of elections thus results in a new bloc whose size is greater than or equal to the largest bloc when considering individual elections, it follows then that VBE has decreased or (at best) stayed the same. $\square$

3.6 Bribery

There is an intuitive relationship between decentralization and bribery. First, it is clear that successful bribery leads to more centralization: in such a case, the entity that acquires the votes of the other players now controls a higher proportion of the total tokens than before. Such a conspicuous observation, however, is challenging to prove in a principled manner. In particular, traditional decentralization metrics, i.e., based on token distribution across accounts, fail to capture this fact; bribed voters, albeit casting votes as instructed by the briber, still technically hold their tokens.

A second, more nuanced observation is that successful bribery must be systemic, i.e., must involve a large number of players, if (and only if) a system is highly decentralized. Intuitively, the more centralized a system is, there is less incentive and opportunity for bribing a large number of players in an election. In this subsection, we show how VBE captures the tight relationship between bribery and decentralization. To do so, we show how our metric fluctuates according to the two scenarios described above, thus yielding a crisp, mathematical characterization of the relationship between decentralization and bribery.

Theorem 3.6. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U_{E,\mathcal{P}}$ the players' utilities across the elections. Assume that an entity successfully bribes players $\hat{\mathcal{P}} \subseteq \mathcal{P}$ in elections E to achieve an outcome of **true**, and denote by $\hat{U}_{E,\mathcal{P}}$ the new utilities of the players in this scenario. Then, it follows that

$$\text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U_{E,\mathcal{P}}) > \text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, \hat{U}_{E,\mathcal{P}}). \quad (9)$$

The proof of this theorem is analogous to that of Theorem 3.4, so we omit it.

Theorem 3.7. Let E be a set of binary elections, $\mathcal{P}$ a set of token holders that participate in such elections, and $U'_{E,\mathcal{P}}$ and $U''_{E,\mathcal{P}}$ two possible utilities of the players across the elections. Assume that an entity needs to bribe other players holding a total of at least n_1 and n_2 tokens to guarantee an outcome of **true** in elections E for the first and second set of utilities, respectively. Then, it follows that

$$n_1 > n_2 \iff \text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U''_{E,\mathcal{P}}) < \text{VBE}_{C_{\epsilon},\min}(E, \mathcal{P}, U'_{E,\mathcal{P}}). \quad (10)$$

Proof. We first make the trivial observation that the minimum number of tokens that must be bought occurs when the bribing entity belongs to the largest voting clique. Thus, let us now compute the relative size of such clique in each of the two scenarios. By the definition of VBE, we get that

$$\begin{aligned}
& -\log_2 \left(\frac{\max_{\mathcal{P}' \in \mathcal{P} / \sim_{U''_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) < -\log_2 \left(\frac{\max_{\mathcal{P}' \in \mathcal{P} / \sim_{U'_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \right) \\
& \iff \frac{\max_{\mathcal{P}' \in \mathcal{P} / \sim_{U'_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} < \frac{\max_{\mathcal{P}' \in \mathcal{P} / \sim_{U''_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}')}{\sum_{P \in \mathcal{P}} \text{tokens}(P)} \\
& \iff C_1 := \max_{\mathcal{P}' \in \mathcal{P} / \sim_{U'_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}') < C_2 := \max_{\mathcal{P}' \in \mathcal{P} / \sim_{U''_{E,\mathcal{P}}}} \text{tokens}(\mathcal{P}').
\end{aligned}$$

Since, for $i \in \{1, 2\}$,

$$\begin{aligned}
n_i &= \frac{\sum_{P \in \mathcal{P}} \text{tokens}(P)}{2} - C_i \\
&\iff C_i = \frac{\sum_{P \in \mathcal{P}} \text{tokens}(P)}{2} - n_i
\end{aligned}$$

it thus follows that

$$\begin{aligned}
& \frac{\sum_{P \in \mathcal{P}} \text{tokens}(P)}{2} - n_1 < \frac{\sum_{P \in \mathcal{P}} \text{tokens}(P)}{2} - n_2 \\
& \iff n_2 < n_1
\end{aligned}$$

as desired. $\square$

The theorem above sheds light on the scale of bribery with respect to the number of tokens that must be corrupted. We can assume additional heuristics to extend this result to other parts of the system. For example, if we assume that the cost of bribery is proportional to the token holdings of the bribed voter, we get that VBE is directly proportional to bribe_P , i.e., the monetary cost of bribery.

3.7 Measuring VBE

In addition to serving as a conceptual tool, variants of VBE can also be measured empirically, and thus quantify the degree of decentralization of a real system. We stress, however, that different instantiations of the VBE framework may require different measurement techniques.

VBE based on voting data. The main component of any measurement strategy for VBE is a way to estimate the utilities of the token holders. This, of course, is guided by the clustering metric being used. Historical voting data is sufficient in our case, since ϵ -TOC assigns clusters based on ordinal utility. If we assume voters are rational, we can extrapolate this from the casted votes: for any player P and election e , it follows that

$$\begin{aligned} \text{vote}_P(e) = \text{true} &\iff \text{util}_P(e, \text{true}) > \epsilon \\ \text{vote}_P(e) = \text{false} &\iff \text{util}_P(e, \text{false}) > \epsilon \\ \text{vote}_P(e) = \perp &\iff |\text{util}_P(e, \text{true})| < \epsilon \end{aligned}$$

Even though we cannot estimate the exact value of $\text{util}_P(e, \text{true})$ based on election outcomes, the equations above are sufficient to use C_ϵ to cluster players.

Data Collection. Even though blockchain-based elections are public, extrapolating relevant voting data for analyzing VBE (and other decentralization metrics) is surprisingly difficult. Indeed, as prior work has also pointed out, “in practice it is not trivial to acquire all governance related information from raw blockchain data” [23]. To aid the analysis and computation of VBE, we thus propose that DAOs publish relevant statistics in a way that is easy to understand and use. Of course, how to best present the data will depend on the concrete instantiation of VBE in question. For example, for ϵ -TOC, the history of casted votes for each individual address is sufficient.

We propose that DAOs choose and specify a variant of VBE to support, which then guides how to present voter data. For ϵ -TOC, DAOs can keep a log mapping all token holders to a list of the elections they were eligible for, denoting their vote (if any). Other variants of VBE may require more detailed information. Feichtinger et al. [23] successfully extrapolated a vast amount of governance-related data from 21 DAOs along multiple axes (albeit noting that it was surprisingly challenging), and open-sourced their data set in the form of “subgraphs” from The Graph protocol [4]. Their work may serve as inspiration for user-friendly ways to present voter data: each DAO could implement a publicly-accessible subgraph of governance data, maintained either by a dedicated set of curators or by the community more broadly.

4 Dark DAOs: Overview

A *Dark DAO* is itself a DAO, but one whose objective is to subvert voting in another DAO or DAOs. Dark DAOs were first introduced in a 2018 blog post [21]. We have shown in Section 3 that as decentralization increases, the cost of a bribery attack rises. Consequently, the need arises for a briber to perform broad coordination, as there is a need to target more users. Thus the threat of Dark DAO deployment increases.

In this section, we briefly explain what Dark DAOs are, giving an informal definition in Section 4.1. We outline their main design principle, *key encumbrance*, in Section 4.2. We explain the various ways in which they can disrupt votes in target DAO Section 4.3.

4.1 Dark DAO Definition

A *Dark DAO* is defined in [21] as a “decentralized cartel that buys on-chain votes opaquely.” Like an ordinary DAO, it’s designed to be trust-minimized: it ensures that a bribe is “fair,” i.e., a bribee receives money iff she votes as directed. Additionally, a Dark DAO is “opaque” in the sense of ensuring that participation is *private*.

Informally, then, a Dark DAO has four essential properties:

1. **Opacity:** Participants in a Dark DAO are indistinguishable on chain from other voters in a DAO. (Consequently, the number of participants is also hidden.)
2. **Guaranteed vote delivery:** A Dark DAO briber is guaranteed that purchased votes will not be manipulated or overridden. A strong variant of this property ensures that a voter will vote as directed. A weak variant ensures that a voter will vote as directed or not vote at all.
3. **Guaranteed briber payment:** A bribee is guaranteed that bribes will be paid according to the Dark DAO’s rules.
4. **Bounded scope:** A bribee who participates in a Dark DAO contributes no resource beyond a committed vote.

4.2 Main tool: Key encumbrance

The main mechanism by which a Dark DAO achieves its properties is *key encumbrance* [29]. Key encumbrance is a form of selective delegation. It enables the Dark DAO to ensure that delegated keys are used to cast the votes to which bribees have committed, but gives the Dark DAO no further control over encumbered keys.

There are two main tools that can enforce such delegation in principle in a way that does not require use of a trusted third party: secure multiparty computation (MPC) [37] and trusted execution environments (TEEs) [34, 33].

TEEs are the more practical, particularly as we demonstrate below that existing hardware-based TEE systems are sufficient to realize Dark DAOs. Such TEEs enable applications to run in an integrity- and confidentiality-protected environment.

A Dark DAO running in a TEE ingests a key sk —serving as a voting credential—that is provided for encumbrance by a bribee. It also takes control of briber cryptocurrency funds. It ensures compliance with the Dark DAO policy, i.e., enforces the properties enumerated in Section 4.1.

4.3 Dark DAO goals

Globally, the goal of a Dark DAO is to subvert voting in a target DAO. There are a number of ways in which it can do this, of which we enumerate several here.

Vote buying: A briber desiring a given election outcome (e.g., a “yes” vote) can simply offer payment for votes toward this outcome—where payments are scaled to the weight associated with a given bribee’s vote (e.g., proportional to her DAO token holdings). Various

forms of conditional payment are also possible, e.g., paying bribes only if the desired outcome is achieved or offering a fixed payment for distribution across the total population of bribees.

We note too that vote buying works not just for systems in which votes are weighted by token holdings, but also “one-vote-per-person” systems, e.g., [36, 42]. In such cases, an encumbered key sk might be a user’s credential in a decentralized identity system, e.g., in Gitcoin Passport or Worldcoin.

Coordinated price manipulation: As noted in [21], it is possible for a Dark DAO to operate without an explicit party distributing bribes. A Dark DAO can instead orchestrate collective action that rewards participants indirectly.

For example, a Dark DAO can orchestrate the following steps among a cabal: (1) Purchase a collective short position in a target asset X ; (2) Vote collectively toward an outcome that causes the price of X to drop; (3) Close the short position at a profit; and (4) Distribute profits among Dark DAO participants. Dark DAO goals can in principle extend beyond voting to other actions as well, such as coordinated attacks on consensus protocols [21] or—if the Dark DAO ingests assets from participants—market manipulation.

Undermining perceived election integrity: The mere presence of a Dark DAO may be enough to cast doubt on a DAO election and call into question whether it is being attacked. DAO opacity conceals the size of a Dark DAO such that even with limited participation, a Dark DAO could have a large psychological impact.

Alternatively, a Dark DAO could selectively reveal (and prove) statistics—e.g., participation of at least 10% of token holdings—that would substantiate the threat it poses.

Exploiting quadratic voting and quadratic funding: Quadratic voting [30] is a mechanism that seeks to limit the influence of whales in determining election outcomes. It weights a given voter’s vote as the square root of her token balance.

Quadratic voting is only enforceable if tokens are assignable to real-world identities. For instance, if votes are weighted as the square-root of token holdings by address, a whale can boost her voting weight by dividing her tokens among multiple accounts.

A Dark DAO can subvert quadratic voting *even when vote is conducted using a secure decentralized identity system*. That is because a Dark DAO can encumber keys not just in a way that enforces voting choices, but *also* use of digital assets.

A whale can thus subvert a quadratic voting scheme as follows. The whale does not just bribe voters to vote for a particular outcome, but also *temporarily deposits some of the whale’s funds with them*. As bribees’ keys are encumbered, the Dark DAO can ensure not just that they vote as directed, but that they will return the whale’s funds.

For example, a whale with 256 tokens can deposit 4 tokens with each of 63 distinct bribees. The result would be an increase in the whale’s voting weight of a factor of $(\sqrt{4 \times 64})/\sqrt{256} = 8!$

A similar attack is possible against quadratic funding [17].

5 Dark DAO Implementation

We implemented a Dark DAO prototype to illustrate that building Dark DAOs is practical. Our implementation is written in a familiar smart contract programming language, Solidity, and demonstrates that developing a Dark DAO need not require any special knowledge of TEEs, thanks to the abstractions provided by the TEE-based Oasis Sapphire blockchain. We demonstrate how Dark DAOs might coordinate bribery on a popular off-chain voting platform, Snapshot⁴, though all DAO voting platforms to our knowledge are susceptible to Dark DAO interference. Our open-source implementation can be found at <https://github.com/DAO-Decentralization/dark-dao>.

In what follows, we first give a background on Oasis Sapphire and Snapshot (Section 5.1) followed by the details of our Dark DAO design (Section 5.2) and possible future design enhancements (Section 5.3). We then discuss the cost of participation (Section 5.4), mitigations against negative impacts of Dark DAOs (Section 5.5) and ethical considerations of building a Dark DAO prototype (Section 5.7).

5.1 Background

Oasis Sapphire Oasis Sapphire is an implementation of the Ethereum Virtual Machine that runs entirely inside a TEE. Assuming the TEE is not broken, Sapphire is able to execute smart contracts whose state is kept private both during execution (in memory) and after execution (in storage). In addition to matching the base implementation of the EVM, Sapphire also includes several built-in precompiled contracts that make it both easy and cheap to perform cryptographic operations pertinent to Dark DAOs: generating entropy and signing messages. These methods are not available in blockchains that do not use TEEs or encrypted computation, since the private key material or entropy would necessarily be leaked to all blockchain nodes.

In its current state, Sapphire does not provide complete privacy to its transactions; the sender and recipient of each transaction and internal transaction are public, and the blockchain’s storage lies outside the TEE. Storage access patterns therefore can leak some information about a contract’s execution or the input data to a transaction [27]. Nonetheless, the privacy Sapphire does have provides enough cover for a Dark DAO: its protection of key material enables key encumbrance, and the confidentiality given to transaction execution helps obscure the relationship between Dark DAO participants and non-participants.

Sapphire has compatibility with many cryptocurrency wallets today, making it a good candidate for hosting a key encumbrance system.

Snapshot Snapshot is an open source, centralized, off-chain voting platform for DAOs. Rather than requiring DAO users to pay the costs of making on-chain voting transactions, votes on Snapshot are submitted as signed messages to the Snapshot website. The website is organized into “spaces,” typically one per DAO, each of which is moderated and/or controlled by a permissioned hierarchy of administrators and moderators of the DAO. At the time of writing, there exist over 28,000 Snapshot spaces [12]. Individual DAOs are free to adjust the algorithm used to calculate the weight of an individual vote, termed as its

⁴<https://snapshot.org>

voting power. How much voting power a particular user gets often is determined by how many DAO tokens he or she is holding on a blockchain at a predetermined block number, and thus a “snapshot” of voting power is taken at that block. For a given DAO proposal, the signed voting messages are collected and, once the voting period is over, are published as receipts to IPFS, a distributed file sharing network. Voters can verify that their votes were included in the proposal’s outcome by checking their voting receipt.

Snapshot also provides a means for *delegating* one’s voting power to another, presumably more active and spirited, voter. One can override his or her delegate’s vote, but generally delegates are chosen based on their advertised values.

5.2 A Key Encumbrance Dark DAO Design

Recall our “guaranteed vote delivery” property—a Dark DAO guarantee to the briber that a purchased vote will actually vote in favor of the briber’s goals. However, many DAO voting systems, including Snapshot, allow voters to change their votes before a proposal has passed. Therefore, the Dark DAO must have exclusive rights for voting (on the user’s behalf) in a target application. At the same time, due to our “bounded scope” property, we cannot grant the Dark DAO complete control over the user’s private key, which may be controlling other, entirely unrelated user assets.

We resolve this tension by designing *key-encumbered wallets*, accounts whose private keys are stored and managed entirely by smart contracts which selectively permit or deny messages from being signed by their owner.

Key-encumbered wallets. Our key-encumbered wallet is powered by a smart contract that runs on Oasis Sapphire. The smart contract generates private-public key-pairs within a TEE using Sapphire’s built-in entropy generation methods, and the private keys are stored encrypted in the smart contract’s storage such that only the smart contract itself can utilize them to sign messages. To create a key-encumbered wallet, one can invoke a “create wallet” function in the smart contract using an external account, typically one that is not encumbered. We emphasize that while the aforementioned external account is the owner of the wallet, it does not have unfettered access to the private keys created and managed by the wallet smart contract.

Owners of key-encumbered wallets can request signatures for messages by issuing read-only calls to the wallet smart contract. These calls are signed by the owner’s external account for authentication.

Encumbrance policies. The owner of a key-encumbered account can invoke a function in the wallet contract to register the account with an *encumbrance policy* smart contract that restricts its ability to sign certain messages. The owner also specifies an expiration time after which the encumbrance policy’s restrictions no longer apply.

All encumbrance policy contracts have the option to restrict which messages or transactions can be signed by the wallet owner, and every message the owner wants to sign is checked with all enrolled encumbrance policies. Any encumbrance policy contract can deny the signing of a message. Those messages which the policy restricts are available for the policy itself to sign on behalf of the wallet owner.

Because encumbrance policies are smart contracts, the encumbrance policy in which an encumbered wallet is enrolled may itself delegate its ability to sign messages on behalf of the owner to other smart contracts. For example, an encumbrance policy that restricts the signing of voting-related messages for a DAO might allow other contracts to restrict the signing of votes for a particular proposal.

Pre-signing attacks Key-encumbered wallet owners can sign an unlimited number of messages prior to enrolling in an encumbrance policy. This creates an opportunity for wallet owners to pre-sign messages they predict will be encumbered by a policy in the future and defeat the encumbrance scheme. For example, if a wallet owner is about to enroll in an encumbrance policy that restricts its ability to sign a message saying “vote for Alice,” the owner could just pre-sign this message before enrolling in the policy. Therefore, encumbrance policies must either be enforced from the moment the wallet is created or work with messages the owner could not have possibly predicted prior to encumbrance, such as those containing a pseudorandom number or the output of a one-way function generated after enrollment.

An alternative key-encumbered wallet design is to record every message that has ever been signed with an encumbered wallet; on enrollment, an encumbrance policy could then check that no previous message breached the encumbrance assumptions of the policy. However, to assess whether a type of message has ever been signed requires each message to be recorded on the same blockchain as the wallet contract, incurring a cost in transaction fees to the wallet owner whenever he or she wishes to sign a message.

Often, these schemes can be combined to produce workable encumbrance policies that would otherwise require enrollment from the moment of wallet creation. If an encumbrance policy can anchor relevant message characteristics to a specific timestamp (e.g., a DAO proposal’s hash to its creation timestamp), it can distinguish whether a particular message could have been signed before the encumbered wallet was enrolled in the policy. All the messages whose timestamps are earlier than the time of enrollment are unrestricted; all those after are eligible for restriction by the policy.

Dark DAOs To create our own Dark DAO based on key-encumbered wallets, we first designed a key encumbrance policy contract that regulates all Snapshot-related messages signed by an enrolled wallet, including votes for DAO proposals. The policy will not allow a key owner to sign a vote directly; instead, the owner must unlock the ability to do so for a particular proposal, after which it can sign any voting message related to that proposal. But rather than unlocking a proposal to sign a vote, an owner may delegate its right to vote to a sub-policy: this is the Dark DAO contract. If the vote is given to a sub-policy, the owner forgoes the ability to sign messages relating to that proposal. This mechanism guarantees to the Dark DAO that a user will not change a vote that it signs on the user’s behalf, so long as the proposal IDs are assumed to be sufficiently unpredictable as to prevent pre-signing attacks. The hierarchical design of key encumbrance enables users to participate in several Dark DAOs at once.

5.3 Flexible Alignment

Rather than controlling votes for a single proposal, Dark DAOs can work to align votes across many proposals. One architecture that achieves this can be set up as follows: the Dark DAO’s victory is contingent on some storage value on the Ethereum mainnet—a property of the victim DAO—changing, such as assigning a privileged role in the DAO to the briber’s address. The Dark DAO does not pay out unless the change is made, and a Merkle proof of the change in the static trie of an Ethereum block header can be given to the Dark DAO at any time; verification makes use of a trusted block header oracle. A bribe is offered inside the Dark DAO’s smart contract on Sapphire. To encourage participants to enter the Dark DAO, the portion of the bribe earmarked for a given entrant is inversely proportional to the amount of voting power already accumulated in the Dark DAO, plus some constant. Dark DAO participants, however, are not aware of their own allocation and will promote others to join in the hopes of a positive outcome that yields them a sizeable portion of the bribe.

Once the condition has been fulfilled, all active Dark DAO participants are paid. The full bribe’s allocation is adjusted based on the alignment of the votes: for example, voting against rest of the Dark DAO incurs a penalty on your bribe payout, with votes for proposals chronologically closer to the moment of victory being weighted more heavily.

5.4 Cost of Participation

Key-encumbered wallets have the potential to undermine one of the core tenets of blockchains: that private keys controlling accounts are owned by individuals rather than groups and that, in the absence of key compromise, messages signed by a particular key reflect the intent of the key’s owner. As we have shown, Dark DAOs form by purchasing the right to perform or prevent the signing of specific messages in encumbered wallets.

We expect that Dark DAOs will not gain much traction initially because of the current lack of standardization for key-encumbered wallets; without encumbrance standards, Dark DAO smart contracts will need to be subject to strict scrutiny before they can be trusted with signing messages. The risk of key compromise via an attack on the TEEs powering the Sapphire network also threatens the funds held in key-encumbered wallets hosted there. Improvements to TEE security and trust will reduce this risk. We do not provide an interface for our key-encumbered wallet contract or for creating Dark DAOs; our implementation merely shows that they are easy to design with today’s software. We have also shown an effective mitigation to Dark DAOs: proofs of complete knowledge.

Nonetheless, through standardization and the design of best practices surrounding encumbrance contracts, some of the risks of participating in Dark DAOs can be reduced.

5.5 Mitigations via Complete Knowledge

An application can prevent access to accounts created by a key encumbrance system by requiring a proof of complete knowledge to be demonstrated for each public key requesting access. Such a proof demonstrates that the associated private key either has been shown or could have been shown, in totality, to an eavesdropper [29]. A key-encumbered wallet can no longer make its encumbrance guarantees if its private key is ever leaked, so correctly

implemented key-encumbered wallets are naturally forbidden from creating valid proofs of complete knowledge.

We are unaware of any protocol currently in existence that incorporates proofs of complete knowledge, so applications themselves will have to check them. A consequence is that applications wishing to escape protocol-level analysis are free to use key encumbrance to create and manage accounts that are indistinguishable from “regular” accounts at the protocol level.

5.6 Limitations

5.7 Ethical Considerations

We have open-sourced the code of our key encumbered wallet and message-based Dark DAO: . We have chosen to do this because, given the current risks of participation, the short-term threat of Dark DAOs is limited. We feel it is important to have a clear demonstration of the practicality of Dark DAOs in the long term. The same code we have written also has beneficial use cases, which we will show in future work. For example, a confidential DAO whose treasury funds are themselves encumbered inside the wallets of its participants would have sidestepped some of the shortcomings of the Constitution DAO, whose public fundraising allowed it to be outbid in a silent auction.

6 Summary Guidance for DAOs

Our VBE framework and the implications we show in Section 3 suggest a number of forms of concrete guidance for DAOs seeking to enforce or improve meaningful decentralization. We discuss them in this section. We summarize our guidance for practitioners in Table 1.

Apathy / inactivity whale and delegation: As we show in Section 3.2, token holders who do not vote—those, in a rational model, with near-zero utility functions—have a centralizing effect. Recall that our term for this group is the *inactivity whale*.

One way to diminish the size of the inactivity whale is through delegation. Intuitively, if tokens associated with the inactivity whale are distributed between at least two delegates in distinct clusters, then they come to represent distinct utility functions—and thus contribute to decentralization.

We show in Section 3.3 that when the inactivity whale is large—with respect to delegates—delegation increases decentralization. (Otherwise, delegation may or may not have this effect.)

Social pressure / voting privacy: There is anecdotal evidence suggesting that social pressure causes voters to align themselves with whales or voting blocs [38]. We may view the effect as a shift in utility function. As we show in Section 3.4, this shift has a centralizing effect.

Social pressure arises because votes are publicly visible. Voting privacy in principle alleviates such pressure and therefore has a decentralizing effect.

<i>Topic</i>	<i>General Guidance</i>	<i>Reason</i>	<i>Relevant result</i>
1. Vote delegation	Given a large inactivity whale, vote delegation tends to increase decentralization.	Delegation (counterintuitively) increases decentralization by diversifying tokens away from a big inactivity whale.	Thm. 3.3
2. Voting privacy	Voting privacy increases decentralization.	Private voting eases social pressure, whose effects are centralizing.	Thm. 3.4
3. Voter bribery	The cost of bribing voters increases with decentralization.	Low alignment of utility functions means a high cost to impose alignment.	Thms. 3.6,3.7
4. Dark DAO risks	Dark DAO risks are likely to increase with decentralization.	As bribery coordination costs grow, Dark DAOs become a more compelling approach to influencing vote outcomes.	Inference from Thm. 3.7
5. Dark DAO feasibility	Dark DAOs are feasible today.	We have shown that existing tools enable effective Dark-DAO deployment. Technical feasibility is unlikely to prove a barrier to their use by adversaries. Complete Knowledge (CK) for voter keys may be a useful countermeasure.	Section 5
6. Identity verification	Weak identity verification increases centralization in quadratic voting.	A whale that can spread tokens across identities amplifies its voting power.	Analysis in Section 4.3
7. Voting slates	Bundling choices into slates (like protocol upgrades that include many voting issues in one package) decreases decentralization.	Bundled choices artificially align otherwise heterogeneous utility functions and/or induce apathy by smoothing out utility functions.	Thm. 3.5
8. Data collection	Careful voting-statistic collection facilitates decentralization measurement.	Lack of systematic collection and publication of detailed voting statistics makes decentralization measurement challenging today.	Data-collection regime suggested in Section 3.7

Table 1: Guidance implied by this paper’s results regarding DAO decentralization.

Snapshot, a popular platform for DAO voting, has recently implemented a form of privacy called *shielded voting* [39]. This form of privacy, however, is only ephemeral: Votes are private when submitted, but revealed at the end of the vote-casting period. So it is

unclear that it can fully address the centralizing effects of social pressure.

End-to-end verifiable voting systems have been proposed in the literature for decades that achieve both voting integrity and confidentiality [14]. How to implement them with token-based weighting is, to the best of our knowledge, though, an open problem.

Voter bribery: Our work shows a relationship between centralization and bribery. In general, bribery causes an increase in centralization, as it has the effect of aligning the utility functions of other players with those of the briber, as we show in Section 3.6.

We also show that as decentralization increases, bribery cost increases. Roughly speaking, increasing diversity in utility functions means increasing cost to align them.

DAOs today are largely centralized [38, 23, 18, 19, 40]. Bribery isn’t especially useful, as whales generally exert strong control and require relatively little coordination to align utility functions into a favorable voting bloc.

One implication of our results is that as DAO decentralization increases, in order for bribery to succeed, it will need to be systemic. DAO designers should therefore recognize large-scale bribery as a future risk.

Dark DAO risks: We hypothesize that the most technically feasible way to implement large-scale bribery is through a Dark DAO.

We have presented in Section 5 the first fully functional private Dark DAO capable of subverting votes on Ethereum. Our architecture leverages the privacy assurances of TEEs in Oasis, but bridges to Ethereum, where most DAOs operate. Our results show that Dark DAOs are technically feasible (and incur low transaction costs) and thus represent a viable future threat.

Dark DAOs pose not just a technical threat, but also a psychological one. Moreover, they can be used not just for direct bribery but also for more subtle attacks. They can, for instance, subvert quadratic voting schemes even when such schemes rely on well-functioning decentralized identity systems.

One possible countermeasure DAO designers may ultimately wish to consider is requiring voting participants to execute complete-knowledge (CK) proofs on their keys [29].

Voting slates: A common trick for passing pieces of legislation that are unpopular or have a narrow base of support is to bundle them together in large, bills. Earmarks are a prime example [16].

This practice may be regarded as a form of utility-function “smoothing”: The utility function of the bill as a whole (for the legislators voting on it) differs from that of its components.

As the practice of bundling proposals / measures has the goal of aligning utility functions, from the standpoint of VBE, it generally has a centralizing effect, as we show in Section 3.5. DAOs may therefore wish to consider limiting the practice and instead explore way to unbundle multi-component proposals.

Data collection: There is no practical way to compute VBE directly—since players’ typically do not express their utility functions. As we discuss in Section 3.7, however, there

are ways to estimate it for a DAO based on voting history. We have found it challenging to collect full voting histories for even popular DAOs. A recommendation for the community is to establish and adhere to standards for archival preservation of DAO voting data.

7 Conclusion and Open Research Questions

We have proposed Voting-Bloc Entropy (VBE) as a new metric for DAO decentralization. VBE measures the entropy of voting blocs. It is in fact a framework into which it is possible to plug any desired method of clustering to identify blocs and any notion of entropy.

Evaluating VBE—instantiated with ϵ -threshold ordinal clustering and min-entropy—we have proven a number of results that held shed light how a number of practices may impact DAO decentralization. We have also shown both in theory and through implementation of a practical system how Dark DAOs pose a potential long-term threat.

Our work gives rise to a number of open research questions. A few deserve particular mention:

- **Privacy:** Our results suggest the potential decentralizing effects of ballot secrecy, i.e., private voting. Existing verifiable end-to-end voting systems implement a one-vote-per-person policy [14]. One open research question is whether token-weighted variants are possible. Additionally, we emphasize in our work the importance of collecting voting data to facilitate VBE estimates. How to harmonize these opposing goals represents a second research challenge.
- **Forking and escape hatches:** DAOs may suffer catastrophic failures, as was famously the case with The DAO [24]. Proposed remedies including forking / splitting, in which a new, quasi-independent or independent DAO is created and escape hatches, which are committee-controlled shutdowns [22]. How their existence and use impact decentralization are unclear and deserves study.
- **VBE impact:** VBE is designed to formalize a view of decentralization in DAOs reflected in the literature and in the views of practitioners. A natural question is what impact high VBE has on decision making in DAOs. Does it correlate with community growth, participation, and financial outcomes in DAOs?

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